NRC7394 SBR Unified Test Guide

Ultra-low power & Long-range Wi-Fi

Ver 1.0
Mar. 4, 2026

NEWRACOM, Inc.

NRC7394 SBR Unified Test Guide
Ultra-low power & Long-range Wi-Fi

Office

Newracom, Inc.
505 Technology Drive, Irvine, CA 92618 USA
http://www.newracom.com

Contents

Overview

This document introduces NEWRACOM's NRC7394 Evaluation kit (EVK). The evaluation kit (EVK) is used to evaluate the performance of the NRC7394 module containing NEWRACOM's IEEE 802.11ah Wi-Fi System on Chip (SoC).

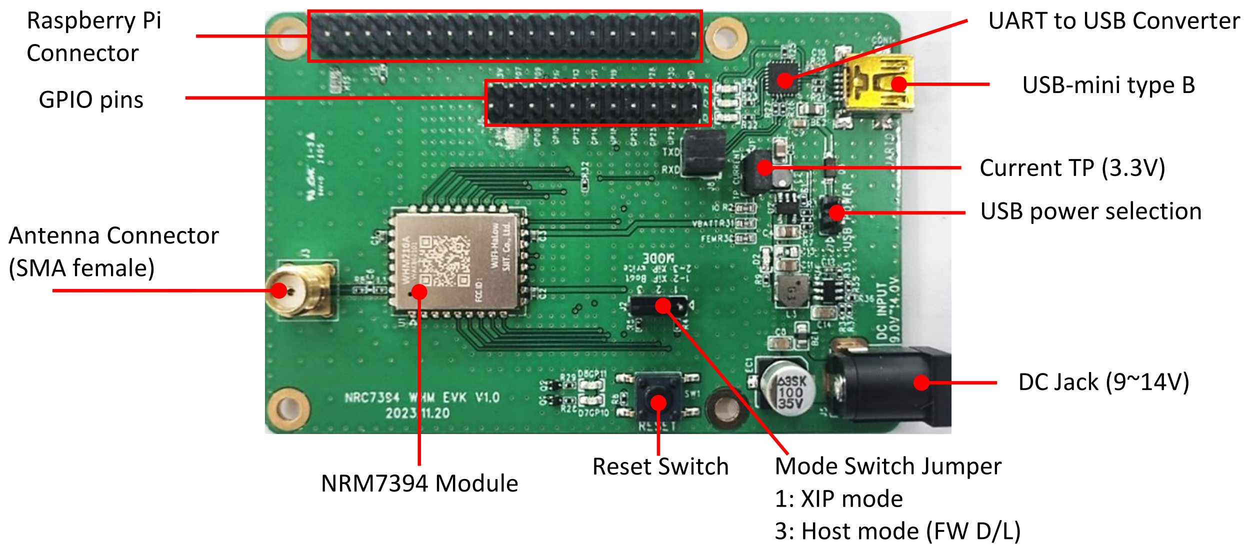


Figure 2.1 WHM evaluation board (Top view)

1.1 HW list

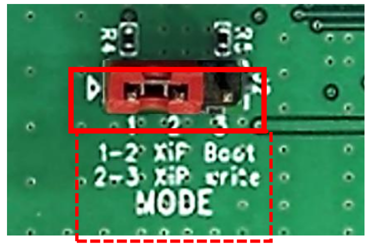
1.1.1 NRC7394 module

NRC7394 module contains IEEE 802.11ah Wi-Fi SoC solution. It also includes a RF front end module (FEM) to increase transmission power up to +27 dBm. Onboard serial flash memory can be used for over-the-air (OTA) software development and with 16KB cache in the NRC7394 supports the execution in place (XIP) feature.

1.1.2 NRC7394 EVB

NRC7394 EVB mainly offers communication interfaces to sensors or an external host.

Configure mode switch to 1-2 (XIP boot) for standalone mode as shown in figures.



WHM EVB

Figure 2.2 Standalone (XIP) mode configuration

The EVB board has a function of step down the DC input power to each voltage source of the module. 3.3V voltage source of the module has a header type test point for easy current measurement.

Please refer to the **Error! Reference source not found.** and Figure 2.3 for the EVB's power structure.

The board has a built-in USB to serial converter and is connected to UART0, so users can check status logs through the onboard USB mini connector.

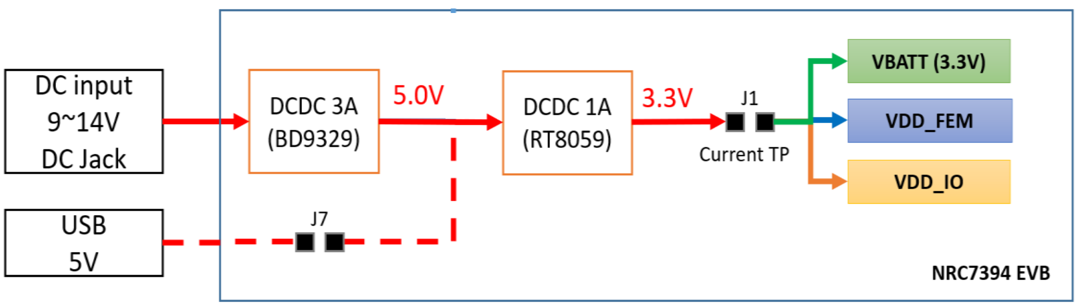


Figure 2.3 WHM EVB Power Structure

1.2 Kit list

NRC7394 EVK includes:

- NRC7394 EVB (NRM7394 module mounted)
- DC 12V (1.5A) power Adaptor
- Antenna (Country frequency)
- USB2.0 mini-B cable
- Raspberry PI4 board (For host mode) with SD card (Linux OS)

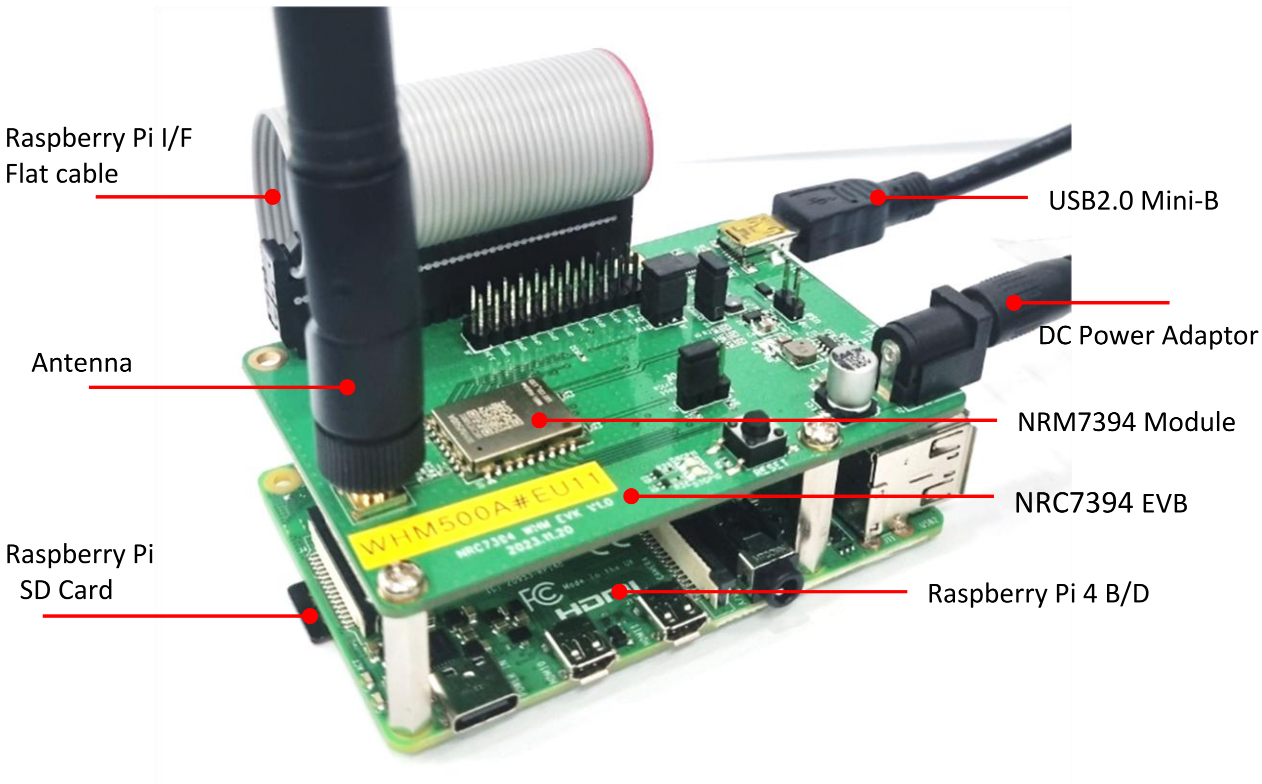


Figure 2.4 NRC7394 hardware set

2 Function test with the SBR FW binary

You can test the FW with a command line interface through a utility program such as TeraTerm. Chapter 2.1 describes how to set up a test environment. Please follow step by step.

2.1 Setup Test Environment

1. Please connect power adapter as below Figure2.1. The reason for using the power adapter is to supply a stable current.
2. We need to connect the PC to the EVB using a UART-USB cable. Please connect the USB cable as below figure 2.1.

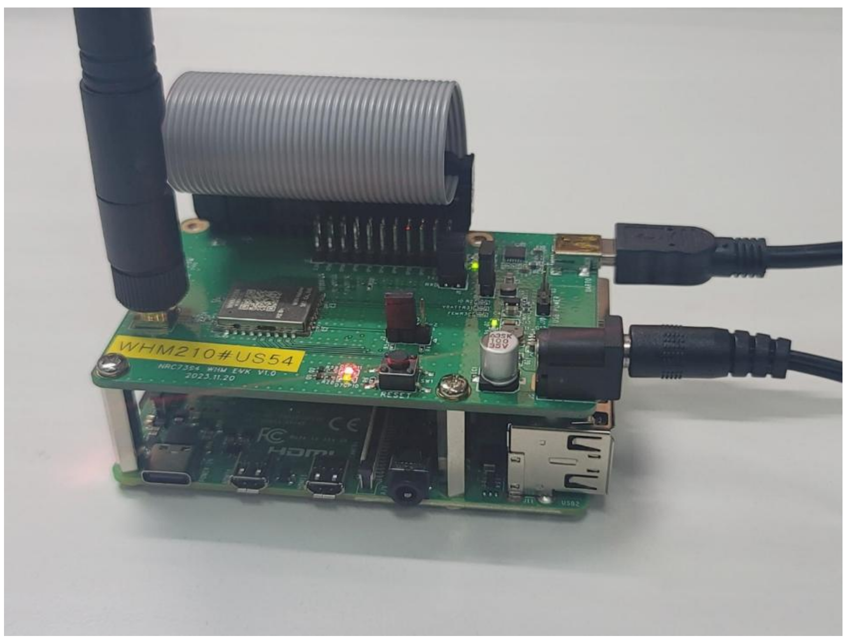


Figure 2.1 Power adaptor and USB cable connection

3. Please check the corresponding COM port number using the Device Manager. The COM port number will be required in the next step.

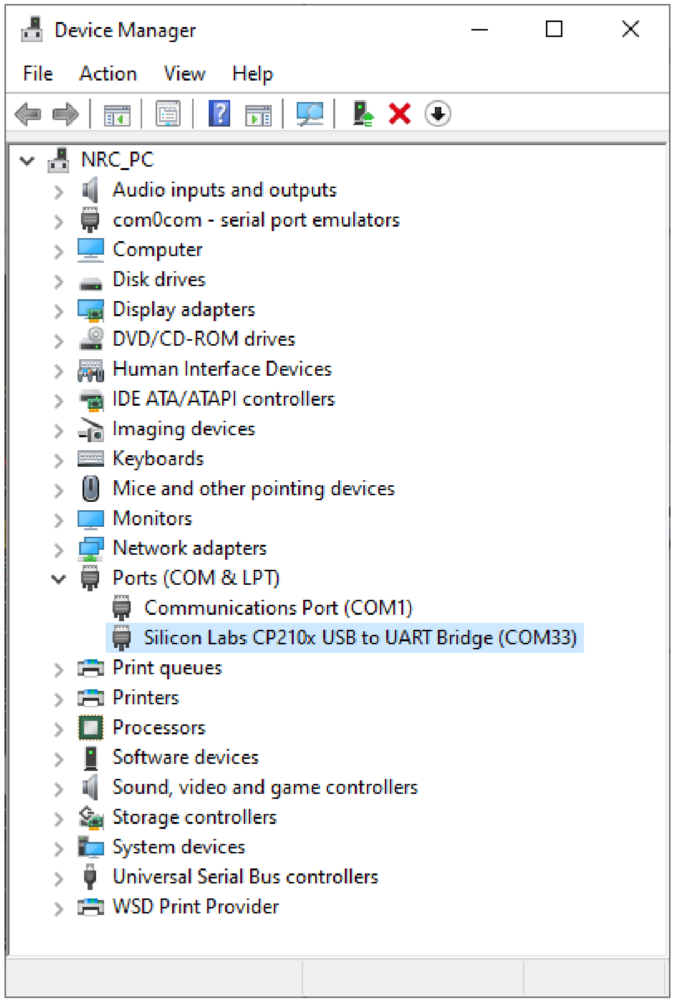


Figure 2.2 COM port in Device Manager

4. Please configure serial port as below. Figure 2.3 show the serial port setru in the TeraTerm. You can download the TeraTerm from https://github.com/TeraTermProject/teraterm/releases/download/v5.2/teraterm-5.2.exe

   A. Speed: 921600
   B. Data : 8 bits
   C. Parity : none
   D. Stop bits : 1bit
   E. Flow control : none

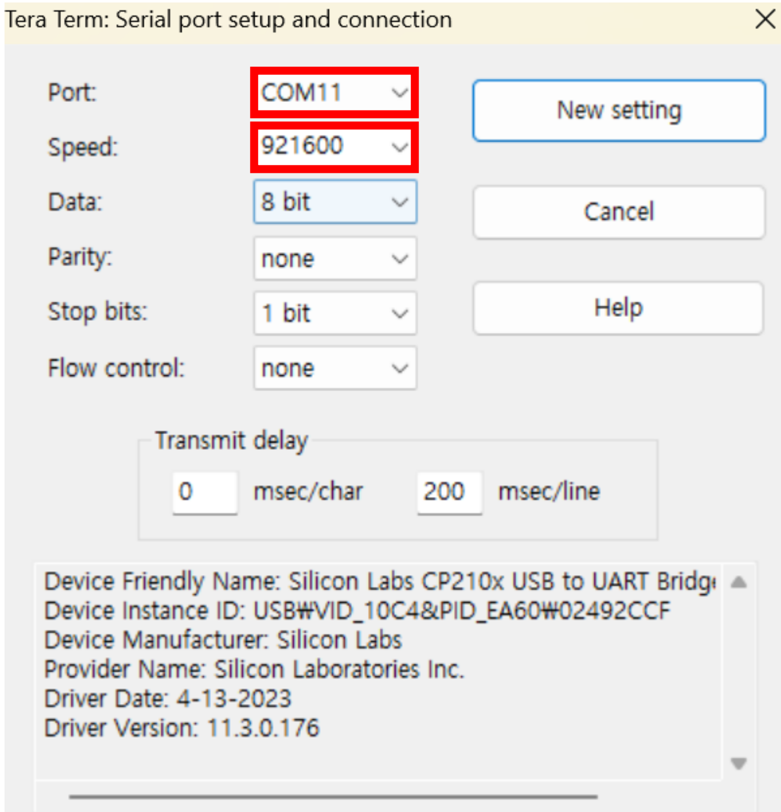


Figure 2.3 Serial port setup in Tera Term

5. Please configure receive new-line character to the AUTO. Figure 2.4 shows Rx new-line configuration of the TeraTerm.

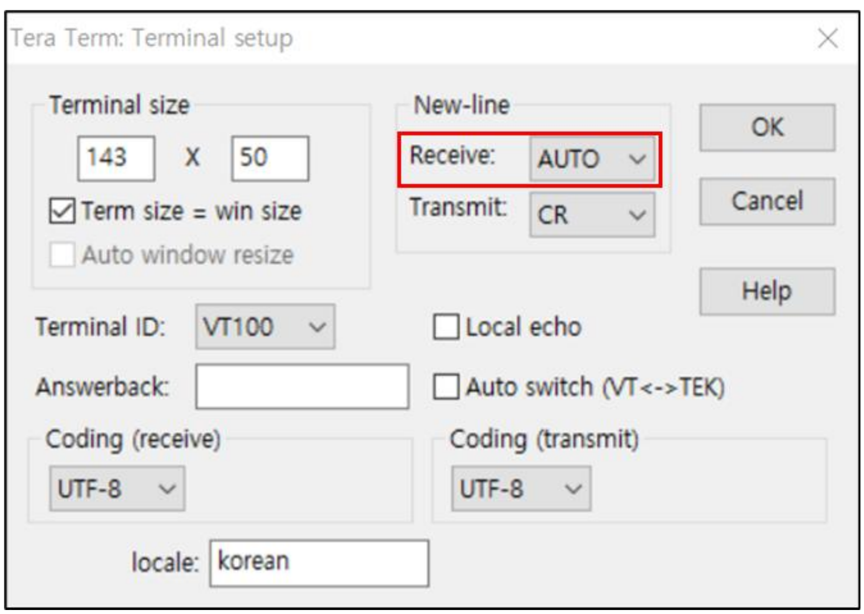
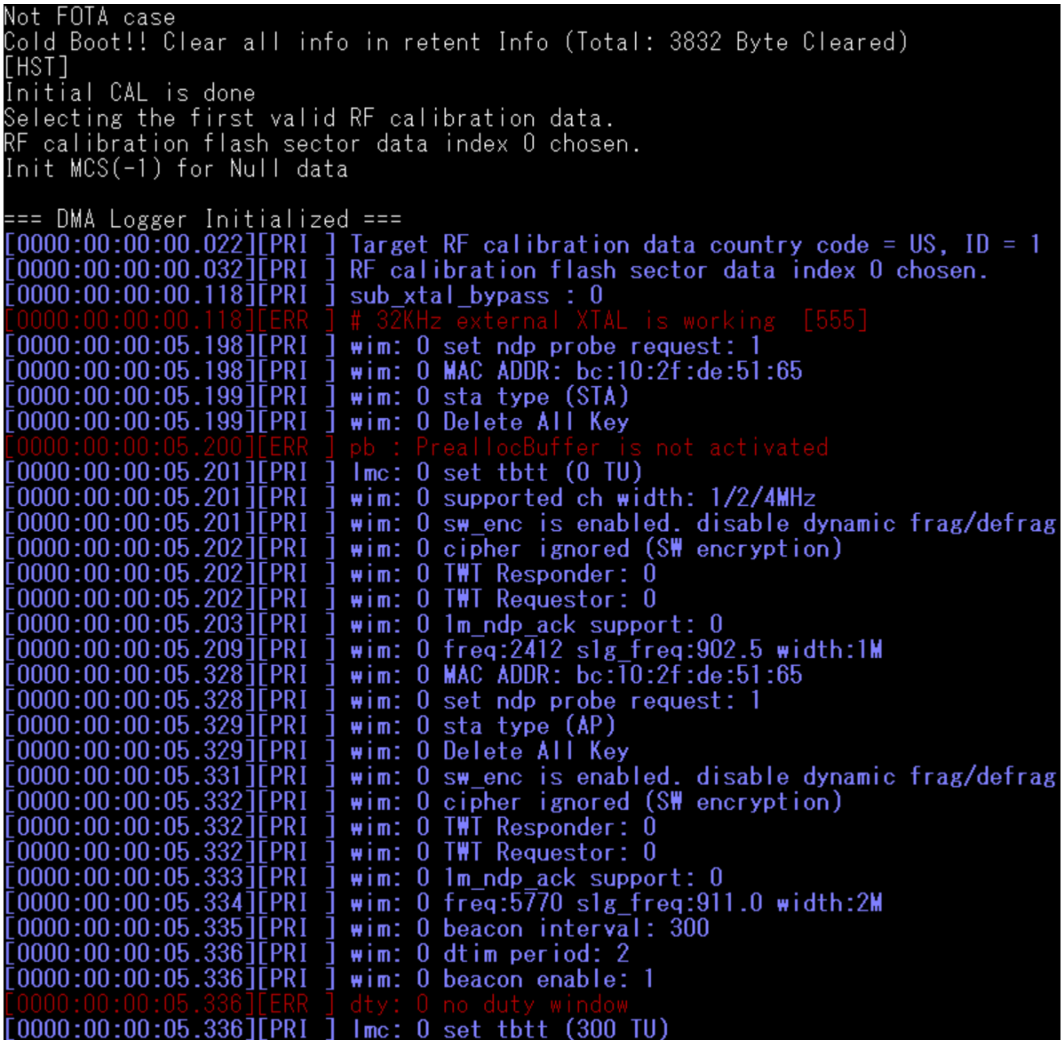


Figure 2.4 Rx New-line configuration in Tera Term

6. Please set beacon_int=300 in the ap_halow_open.conf file.
   On the host, run the following command: ./start_modular.py 1 0 US
   Finally, you can see the NRC7394 command-line interface in Figure2.5.



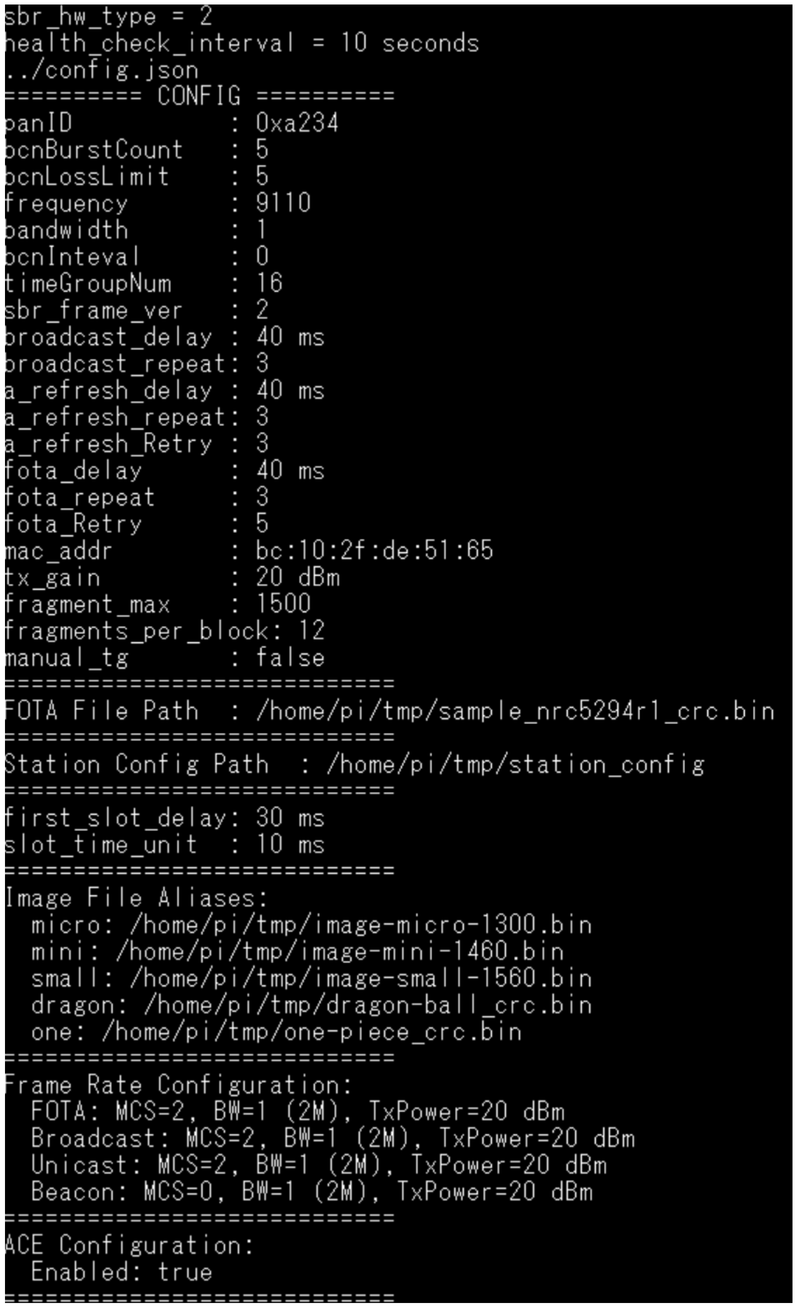
```
Not FOTA case
Cold Boot!! Clear all info in retent Info (Total: 3832 Byte Cleared)
[HST]
Initial CAL is done
Selecting the first valid RF calibration data.
RF calibration flash sector data index 0 chosen.
Init MCS(-1) for Null data

=== DMA Logger Initialized ===
[0000:00:00:00.022][PRI ] Target RF calibration data country code = US, ID = 1
[0000:00:00:00.032][PRI ] RF calibration flash sector data index 0 chosen.
[0000:00:00:00.118][PRI ] sub_xtal_bypass : 0
[0000:00:00:00.118][ERR ] # 32KHz external XTAL is working  [555]
[0000:00:00:05.198][PRI ] wim: 0 set ndp probe request: 1
[0000:00:00:05.198][PRI ] wim: 0 MAC ADDR: bc:10:2f:de:51:65
[0000:00:00:05.199][PRI ] wim: 0 sta type (STA)
[0000:00:00:05.199][PRI ] wim: 0 Delete All Key
[0000:00:00:05.200][ERR ] pb : PreallocBuffer is not activated
[0000:00:00:05.201][PRI ] lmc: 0 set tbtt (0 TU)
[0000:00:00:05.201][PRI ] wim: 0 supported ch width: 1/2/4MHz
[0000:00:00:05.201][PRI ] wim: 0 sw_enc is enabled. disable dynamic frag/defrag
[0000:00:00:05.202][PRI ] wim: 0 cipher ignored (SW encryption)
[0000:00:00:05.202][PRI ] wim: 0 TWT Responder: 0
[0000:00:00:05.202][PRI ] wim: 0 TWT Requestor: 0
[0000:00:00:05.203][PRI ] wim: 0 1m_ndp_ack support: 0
[0000:00:00:05.209][PRI ] wim: 0 freq:2412 s1g_freq:902.5 width:1M
[0000:00:00:05.328][PRI ] wim: 0 MAC ADDR: bc:10:2f:de:51:65
[0000:00:00:05.328][PRI ] wim: 0 set ndp probe request: 1
[0000:00:00:05.329][PRI ] wim: 0 sta type (AP)
[0000:00:00:05.329][PRI ] wim: 0 Delete All Key
[0000:00:00:05.331][PRI ] wim: 0 sw_enc is enabled. disable dynamic frag/defrag
[0000:00:00:05.332][PRI ] wim: 0 cipher ignored (SW encryption)
[0000:00:00:05.332][PRI ] wim: 0 TWT Responder: 0
[0000:00:00:05.332][PRI ] wim: 0 TWT Requestor: 0
[0000:00:00:05.333][PRI ] wim: 0 1m_ndp_ack support: 0
[0000:00:00:05.334][PRI ] wim: 0 freq:5770 s1g_freq:911.0 width:2M
[0000:00:00:05.335][PRI ] wim: 0 beacon interval: 300
[0000:00:00:05.336][PRI ] wim: 0 dtim period: 2
[0000:00:00:05.336][PRI ] wim: 0 beacon enable: 1
[0000:00:00:05.336][ERR ] dty: 0 no duty window
[0000:00:00:05.336][PRI ] lmc: 0 set tbtt (300 TU)
```

Figure 2.5 Command Line interface

2.2 Test application

1. Please configure the **Config.json** file, ensure that the file paths are set correctly, and make sure the **mac_addr** matches the **macaddr** defined in start_modular.py. Please also ensure that the **frequency** value is set to the same value as specified in the ap_halow_open.conf file.
2. On the host, run the following command: ./test -s2 -c ../config.json
   A screenshot of the commands entered is shown in Figure 2.6.



```
sbr_hw_type = 2
health_check_interval = 10 seconds
../config.json
========== CONFIG ==========
panID            : 0xa234
bcnBurstCount    : 5
bcnLossLimit     : 5
frequency        : 9110
bandwidth        : 1
bcnInteval       : 0
timeGroupNum     : 16
sbr_frame_ver    : 2
broadcast_delay  : 40 ms
broadcast_repeat: 3
a_refresh_delay  : 40 ms
a_refresh_repeat: 3
a_refresh_Retry  : 3
fota_delay       : 40 ms
fota_repeat      : 3
fota_Retry       : 5
mac_addr         : bc:10:2f:de:51:65
tx_gain          : 20 dBm
fragment_max     : 1500
fragments_per_block: 12
manual_tg        : false
============================
FOTA File Path   : /home/pi/tmp/sample_nrc5294r1_crc.bin
============================
Station Config Path  : /home/pi/tmp/station_config
============================
first_slot_delay: 30 ms
slot_time_unit   : 10 ms
============================
Image File Aliases:
  micro: /home/pi/tmp/image-micro-1300.bin
  mini: /home/pi/tmp/image-mini-1460.bin
  small: /home/pi/tmp/image-small-1560.bin
  dragon: /home/pi/tmp/dragon-ball_crc.bin
  one: /home/pi/tmp/one-piece_crc.bin
============================
Frame Rate Configuration:
  FOTA: MCS=2, BW=1 (2M), TxPower=20 dBm
  Broadcast: MCS=2, BW=1 (2M), TxPower=20 dBm
  Unicast: MCS=2, BW=1 (2M), TxPower=20 dBm
  Beacon: MCS=0, BW=1 (2M), TxPower=20 dBm
============================
ACE Configuration:
  Enabled: true
============================
```

Figure 2.6 Logs with the commands are entered

3. run the following command: ap-config

```
>> ap-config
success: update sbr id
success: sbr hw version (version=2)
success: modem tx gain
success: ap address
success: group wakeup config
success: frame rate config (FOTA MCS=2 BW=1, Broadcast MCS=2 BW=1, Unicast MCS=2 BW=1)
success: ap config
success: ACE enable
```

Figure 2.7 Logs with the commands are entered

```
[0000:00:33:35.238][PRI ] ----- Config AP ------
[0000:00:33:35.238][PRI ] macHeader: T=119, B=(0/5), FC=(0,0,0,0,1), P=0xA234
[0000:00:33:35.239][PRI ] bcnMode: M=0, V=0, R=0
[0000:00:33:35.239][PRI ] bcnMacCfg: I=0, G=0, C=0
[0000:00:33:35.239][PRI ] syncOption: L=5, R=0, S=0, B=0, F=0, E=0
[0000:00:33:35.240][PRI ] etcOption: U=0, P=0, R=0
[0000:00:33:35.240][PRI ] gwCapa: A=0, C=0
[0000:00:33:35.240][PRI ] tagPageLock: P=0, L=0, R=0
[0000:00:33:35.240][PRI ] tagBattIcon: M=0, L=0
[0000:00:33:35.241][PRI ] SBR ID: 0x00000001
[0000:00:33:35.241][PRI ] Freq: 9110, BW:1
[0000:00:33:35.241][PRI ] UTC: 1772592785, DIV:0, RX-TO:0
[0000:00:33:35.242][PRI ] Duration: 1560us
[0000:00:33:35.242][PRI ] MAC Address: bc:10:2f:de:51:65
[0000:00:33:35.242][PRI ] ---------------------
```

Figure 2.8 Logs are shown in the NRC7394 console

4. When the **NRC5294 Tag** starts and connects, it is added to the STA list, and logs are displayed as follows.

```
>> sta list
Group[0] (Stations: 0)
Group[1] (Stations: 0)
Group[2] (Stations: 0)
Group[3] (Stations: 0)
Group[4] (Stations: 0)
Group[5] (Stations: 1)
  ID: 0x00000002, 40:16:44:3E:51:1C, SG: 12, Ver: 0x01, cnt: 2, nb: 0, OK, 2026-03-04 02:59:02
Group[6] (Stations: 0)
Group[7] (Stations: 0)
Group[8] (Stations: 0)
Group[9] (Stations: 0)
Group[10] (Stations: 0)
Group[11] (Stations: 1)
  ID: 0x00000003, 40:16:EE:CF:E7:B1, SG: 1, Ver: 0x01, cnt: 1, nb: 0, OK, 2026-03-04 03:01:19
Group[12] (Stations: 0)
Group[13] (Stations: 0)
Group[14] (Stations: 0)
Group[15] <-- Current (Stations: 0)

========================================
Total Stations: 2
========================================
```

Figure 2.9 Logs with the “sta list” commands are entered

```
test[26534]: PROTOCOL-RX: msgType=0x79
test[26534]: IPC: IPC_STA_FIND-mac=40:16:ee:cf:e7:b1
test[26534]: IPC: IPC_STA_FIND-station not found
test[26534]: Allocated ID: 0x00000003
test[26534]: PROTOCOL-EVENT: received type=4002, length=504
test[26534]: received EVENT: immediate report
test[26534]: Immediate Report:
test[26534]:   n_pkt: 1, reserved: 0
test[26534]: Buffer[0]: ID=0x0003, Status=TX-OK
test[26534]: Station FW Version: 0x01
test[26534]: Station Added: ID:0x00000003, conn_cnt=1
```

Figure 2.10 Logs are displayed in mylog

```
[0000:00:00:09.095] ########################################
[0000:00:00:09.095] # Version : vMaster
[0000:00:00:09.095] # Build   : Mar  4 2026 17:26:08
[0000:00:00:09.095] ########################################
[0000:00:00:09.095] SPI DMA: Started
[0000:00:00:09.096] Start MAC Process
[0000:00:00:09.096] Start App Process
[0000:00:00:09.097] Start Act Process

[0000:00:00:09.097] ACT: Try Activation Request 1/3
[0000:00:00:09.097] APP: Best AP frequency=9110, bw=1
[0000:00:00:09.106] APP: Boot Reason=BOOT_ON_RTC (0x20)
[0000:00:00:09.107] Temp Value:23
[0000:00:00:09.107] VBATT Value:3279
[0000:00:00:09.108] APP-RX: [ACT_RSP], SNR=32, RSSI=-7, MCS=2, SIZE=65, SEQ=2, RETRY=0, bc:10:2f:de:51:65
[0000:00:00:09.109] ACT: Received ActRsp from bc:10:2f:de:51:65
[0000:00:41:56.736] APP: Time Sync 8896207us -> 2457750706us
[0000:00:41:56.737] APP: activation success event
[0000:00:41:56.737] APP: SBR RX Tracking Disabled
[0000:00:41:56.737] msgType=122, panID=0x0000A234, timestamp=2457748359us
[0000:00:41:56.738] FC(E128=0 E256=0, LBTN=0, SBRID=0x00000001)
[0000:00:41:56.738] result=1, jobNumber=0, isShutdown=0
[0000:00:41:56.739] allocTagSBRID=3, slot_time_unit=10, first_slot_delay=30, allocGroup=11, allocSubGroup=1
PSM: Next Wakeup=-1ms
```

Figure 2.11 Logs are shown in the NRC5294 console

5. run the following command: sta wakeup dragon 3
   The dragon-ball_crc.bin image is updated to the **NRC5294 Tag** with ID 3.

```
test[26534]: Schedule Data: group[11], buffer_id=0x0006, sbr_id=0x00000003, len:1500
test[26534]: Schedule Data: group[11], buffer_id=0x0007, sbr_id=0x00000003, len:1500
test[26534]: Schedule Data: group[11], buffer_id=0x0008, sbr_id=0x00000003, len:1500
test[26534]: Schedule Data: group[11], buffer_id=0x0009, sbr_id=0x00000003, len:620
kernel:  nrc-mcp: Info MCP: Processing data H2F len=1508
kernel:  nrc-mcp: Info MCP: TX via MCP queue, subtype=33, len=1508, hif_header_included=0
kernel:  nrc-mcp: Info MCP: Processing data H2F len=1508
kernel:  nrc-mcp: Info MCP: TX via MCP queue, subtype=33, len=1508, hif_header_included=0
kernel:  nrc-mcp: Info MCP: Processing data H2F len=1508
kernel:  nrc-mcp: Info MCP: TX via MCP queue, subtype=33, len=1508, hif_header_included=0
kernel:  nrc-mcp: Info MCP: Processing data H2F len=628
kernel:  nrc-mcp: Info MCP: TX via MCP queue, subtype=33, len=628, hif_header_included=0
test[26534]: PROTOCOL-EVENT: received type=4001, length=96
test[26534]: received EVENT: schedule report
test[26534]: === SCHEDULE REPORT DETAILS ===
test[26534]: Group ID: 11, Number of Reports: 1
test[26534]: --- Report[0] ---
test[26534]:   RxDataType: 3 (report(o),req(o))
test[26534]:   RxRssi: 24 dBm
test[26534]:   TagAddr: 40:16:EE:CF:E7:B1
test[26534]:   TagStatus:
test[26534]:     Battery: 32 (100mV units)
test[26534]:     Temperature: 0
test[26534]:     LastDisplayStatus: 0
test[26534]:   Button Counts:
test[26534]:     Btn1: 0, Btn2: 0, Btn3: 0, Btn4: 0
test[26534]:   Long Button Counts:
test[26534]:     Long_Btn1: 0, Long_Btn2: 0, Long_Btn3: 0, Long_Btn4: 0
test[26534]:   Button Event: btn1=0 btn2=0 btn3=0 btn4=0 long_btn1=0 long_btn2=0 long_btn3=0 lon

test[26534]:   TagMsgType: 0x00
test[26534]:   TranStatus: 0 (unknown)
test[26534]:   InquiryPage: 0
test[26534]:   JobNumInt: 20
test[26534]:   Block Info:
test[26534]:     TotalBlock: 1, CurrentBlock: 0
test[26534]:   BlinkSeqNum: 0xFF
test[26534]:   TxSeqNum: [00 00 00 00 00 00 00 00 00]
test[26534]:   ControlID: [00 00 00 00 00 00 00]
test[26534]: === END SCHEDULE REPORT ===
test[26534]: HealthMonitor: Current status: HEALTHY -> HEALTHY (driver_fail:0/3, fw_fail:0/3)
```

Figure 2.12 Logs are displayed in mylog

```
[0000:00:52:04.065] ########################################
[0000:00:52:04.065] # Version : vMaster
[0000:00:52:04.065] # Build   : Mar  4 2026 17:26:08
[0000:00:52:04.066] ########################################
[0000:00:52:04.066] SPI DMA: Started
[0000:00:52:04.066] Start MAC Process
[0000:00:52:04.067] Start App Process
[0000:00:52:04.067] Start Act Process

[0000:00:52:04.068] APP: Boot Reason=BOOT_ON_SBR_WAKEUP (0x2)
[0000:00:52:04.069] Temp Value:27
[0000:00:52:04.069] VBATT Value:3288
[0000:00:52:04.070] APP: Best AP frequency=9110, bw=1
[0000:00:52:04.071] MULTI-BLK: Sending multi-block inquiry request, battery=32
[0000:00:52:04.102] APP-RX: [INQ_RSP], SNR=34, RSSI=-7, MCS=2, SIZE=1500, SEQ=5, RETRY=0, bc:10:2f:de:51:65
[0000:00:52:04.109] APP-RX: [INQ_RSP], SNR=32, RSSI=-7, MCS=2, SIZE=1500, SEQ=6, RETRY=0, bc:10:2f:de:51:65
[0000:00:52:04.117] APP-RX: [INQ_RSP], SNR=33, RSSI=-7, MCS=2, SIZE=1500, SEQ=7, RETRY=0, bc:10:2f:de:51:65
[0000:00:52:04.121] APP-RX: [INQ_RSP], SNR=32, RSSI=-7, MCS=2, SIZE=620, SEQ=8, RETRY=0, bc:10:2f:de:51:65
[0000:00:52:04.122] MULTI-BLK: Trying to send report

[0000:00:52:04.128] APP: Time Sync 3050906474us -> 3050906544us
[0000:00:52:04.129] APP: multi block done!!!!
[0000:00:52:04.129] SPI DMA: Requesting 4744 bytes
[0000:00:52:04.129] SPI DMA[0]: Sending 4095 bytes at offset 0
[0000:00:52:04.129] SPI DMA[0]: Sending 649 bytes at offset 4095
[0000:00:52:04.130] SPI DMA: Delay 1 Sec
[0000:00:52:04.343] APP-RX: [BEACON_], SNR=31, RSSI=-7, MCS=0, SIZE=103, SEQ=3525, RETRY=1, bc:10:2f:de:51:65
[0000:00:52:04.653] APP-RX: [BEACON_], SNR=34, RSSI=-7, MCS=0, SIZE=103, SEQ=3526, RETRY=1, bc:10:2f:de:51:65
[0000:00:52:04.968] APP-RX: [BEACON_], SNR=33, RSSI=-7, MCS=0, SIZE=103, SEQ=3527, RETRY=1, bc:10:2f:de:51:65
[0000:00:52:05.154] SPI DMA[1]: Sending 4095 bytes at offset 0
[0000:00:52:05.155] SPI DMA[1]: Sending 649 bytes at offset 4095
[0000:00:52:05.155] APP: spi dma done
PSM: Next Wakeup=-1ms
```

Figure 2.13 Logs are shown in the NRC5294 console

2.3 Test application commands

Command	Args	Description
ace-enable	ace-enable <0|1>	Enable or disable ACE (Activation Control Entity).
auto-refresh	auto-refresh <start|stop> [period] [utc_time] Example: auto-refresh start auto-refresh start 120 auto-refresh start 60 1709568000 auto-refresh stop	Start or stop automatic periodic refresh.
cspi		Send a CSPI test packet with a corrupted checksum for firmware error-handling verification.
update-sbrid	mcp-flash-write <filename> <info|boot> <sbr_id> Example: mcp-flash-write device_info.json info 0x02 mcp-flash-write config.json info 5	Generate a new random SBR ID and update it on the modem.
calc	calc frame <bw> <mcs> <length> Example: calc frame 1 7 1024	Calculate wireless packet duration.
test	test <n_station>	Add synthetic test stations for debugging purposes.
mcp-specific		Send a raw MCP vendor-specific command to the modem.
fota	fota <start|stop> [file] [tag_type] [fw_ver] [chip_ver] Examples: fota start fota start /path/to/firmware.bin fota start /path/to/firmware.bin 0x01 0x02 0x01 fota stop	Start or stop Firmware Over-The-Air (FOTA) update.
utc		Update the modem's UTC time to the current system time.
firmware	firmware [name]	Load a firmware binary into the modem.
ap-config		Configure the Access Point with all settings.

mcp-flash-write	mcp-flash-write <filename> <info|boot> <sbr_id>	Write a JSON configuration file to a station's flash memory via MCP specific frame.
help		Display a list of all available commands.
sta-config	sta-config list sta-config set	Manage station JSON configuration files.
log	log <module> <level> Examples: log mac debug	Configure driver logging level for a specific module.
credit		Display the driver credit (TX/RX slot and data counters).
ping	ping <target> Examples: ping driver ping firmware ping both	Test driver and/or firmware connectivity.
image	image list	Display image file alias-to-path mappings.
group	group list group wakeup <image_alias> <group_index|all> Examples: group wakeup dragon 0 group wakeup dragon all	List groups or wakeup all stations in a group.
subgroup	subgroup wakeup <image_alias> <group_id> <sbr_id> [sbr_id ...] Examples: subgroup wakeup dragon 0 0x02 0x03 subgroup wakeup dragon 1 2-10	Wakeup specific stations within a group (subgroup wakeup).
exit		Exit the CLI application.
ap-config-down		Bring the Access Point down (deactivate).
sta	sta list sta clear sta wakeup <image_alias> <sbr_id> [sbr_id ...] Examples: sta wakeup dragon 0x02 sta wakeup dragon 2 3 4 5 sta wakeup dragon 2-77	List, clear, or wakeup stations.

3 Abbreviations and acronyms

Abbreviations Acronyms	Definition
SBR	Standby Radio
AP	Access Point
EVB	Evaluation Board
EVK	Evaluation Kit
FEM	Front End Module
GPIO	General Purpose Input Output
IEEE	Institute of Electrical and Electronics Engineers
OTA	Over-the-Air
RPi3	Raspberry Pi 3
SDK	Software Development Kit
SoC	System on Chip
STA	Station
UART	Universal Asynchronous Receive Transmitter
USB	Universal Serial Bus
XIP	eXecute In Place
TBTT	Target Beacon Transmit Time

4 Revision history

Revision No	Date	Comments
Ver 1.0	Mar/4/2025	Initial version